Auditing Surface-Order Approaches to GRH: Detailed Proofs and Research Interfaces

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Research draft. GRH is not proved here.

Abstract

We audit proposed surface-order and Hilbert–Pólya arguments for GRH. We prove obstructions to the proposed prime/geodesic identification, integral cyclic induction, absolute prime-tail estimates and determinant preservation, and retain a conditional spectral closure criterion. We construct convergent arithmetic approximants with explicit errors and analyze coefficient-defined Jensen metrics. From the explicit scalar local factors of a separate exponent argument, we extend an Euler-correction region and its principal-residue interface, and isolate a shifted infinite-order angular Hecke factor. Under the stated classical analytic inputs, its zeros obstruct extending the unchanged correction's nonvanishing criterion below 7/12. A pinned Lean 4 companion verifies selected arithmetic, complex-analytic, matrix and infinite-product components, with its remaining dependencies specified. These results do not establish GRH; the full arithmetic spectral bridge and the required mean-square and contour estimates remain unresolved.

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1 Scope and conventions

For a primitive Dirichlet character χ of conductor q, write

$$L(s,\chi)=\sum_{n\geq 1}\frac{\chi(n)}{n^s},\qquad \Lambda(s,\chi)=\left(\frac{q}{\pi}\right)^{(s+a)/2}\Gamma\left(\frac{s+a}{2}\right)L(s,\chi),\quad a=\frac{1-\chi(-1)}{2}.$$

The series and Euler product initially apply for $\operatorname{Re} s>1$. Our target is Dirichlet GRH: every zero in $0<\operatorname{Re} s<1$ has real part $1/2$. For $q=1$, use the entire function $\xi(s)=\frac{1}{2}s(s-1)\Lambda(s,1)$ when an entire target is required. Finite-order Hecke characters and general Artin representations require separate constructions; a theorem for one family is not silently a theorem for another.

The supplied surface-order manuscripts [1, 2] propose arithmetic transfer operators and self-adjoint approximants. We isolate valid closure logic and prove obstructions to particular claimed constructions. These obstructions refute intermediate statements, not GRH. The separate `math` repository [3] includes claimed zero-free regions strictly weaker than GRH. Such a region cannot replace the missing critical-line argument.

2 A corrected conditional closure theorem

Theorem 2.1 (Spectral approximation criterion)**.** *Let $U\subset\mathbb{C}$ be connected and open, let f be holomorphic on U and not identically zero, and let*

$$F_N(s)=e^{a_N+b_N s}\det(sI-(\tfrac{1}{2}I+iH_N)),\qquad H_N=H_N^*\in M_{d_N}(\mathbb{C}).$$

If $F_N\to f$ locally uniformly on U, then every zero of f in U has real part $1/2$.

Proof. The eigenvalues $\lambda_{j,N}$ of H_N are real. Thus the zeros of F_N are $1/2+i\lambda_{j,N}$, with algebraic multiplicities; the exponential factor never vanishes. If $f(z)=0$ off that line, choose a closed disk $D\subset U$ about z disjoint from the line, with f nonzero on its boundary. Such a disk exists because the zeros of a nonzero holomorphic function on a connected domain are isolated. Let $m=\min_{\partial D}|f|>0$. Local uniform convergence gives $|F_N-f|<m$ on ∂D for sufficiently large N. Rouché's theorem forces F_N and f to have the same positive number of zeros inside D, contradicting the zero location of F_N. □

Remark 2.2. This is a conditional criterion, not a construction. Self-adjointness alone establishes only the zero location of the approximants. The difficult arithmetic statement is convergence to the particular completed L-function. Pointwise convergence, weak operator convergence, and agreement of finitely many zeros do not supply the displayed hypothesis.

Proposition 2.3 (A usable continuation interface)**.** *Suppose U is a connected domain, the entire functions F_N are locally uniformly bounded on U, and they converge pointwise on a nonempty open subset $V\subset U$ to a holomorphic function f defined on U. Then $F_N\to f$ locally uniformly on U.*

Proof. Montel's theorem makes the family normal. Every subsequence has a further subsequence converging locally uniformly to a holomorphic function g. On V, pointwise convergence gives $g=f$, so the identity theorem gives $g=f$ on U. If the original sequence failed to converge locally uniformly, a compact set, positive error, and violating subsequence would contradict this subsequence property. □

A proposed analytic continuation proof must actually provide the uniform bounds and a genuine overlap V. The disjoint domains $\operatorname{Re} s>1$ and $0<\operatorname{Re} s<1$ are not an overlap.

3 Arithmetic obstructions to the claimed completion

Theorem 3.1 (No integral prime-power geodesic norms)**.** *Let $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ be hyperbolic. Its expanding eigenvalue in absolute value is $u > 1$, and its hyperbolic norm is $N(\gamma) = u^2$. Then $N(\gamma)$ is not an integer greater than one. In particular no closed geodesic arising from such a matrix has length $k \log p$ for a prime p and integer $k \geq 1$.*

Proof. Put $t = |\operatorname{tr} \gamma| \in \mathbb{Z}$, so $t = u + u^{-1}$. If $u^2 = n \in \mathbb{Z}$, $n > 1$, squaring gives

$$t^2 = n + 2 + \frac{1}{n}.$$

The left-hand side and $n + 2$ are integers, so $1/n$ is an integer, contradicting $0 < 1/n < 1$. Since length is $\log N(\gamma)$, length $k \log p$ would imply $N(\gamma) = p^k$, already excluded. □

The statement applies to every subgroup of $\mathrm{SL}_2(\mathbb{Z})$, hence to congruence surfaces. The prime geodesic theorem is an asymptotic counting analogy; it cannot provide the termwise identification claimed in the completion section of [1]. Hecke correspondences of determinant p must not be identified with hyperbolic deck transformations of determinant one.

Theorem 3.2 (Failure of integral cyclic induction)**.** *The trivial character of $G = C_2 \times C_2$ is not an integer linear combination of characters induced from one-dimensional characters of cyclic subgroups of G.*

Proof. Every cyclic subgroup has order one or two. The degree of any such induced character is its index, namely four or two. Evaluation at the identity sends every integer linear combination to an even integer, whereas the trivial character has degree one. □

This distinguishes rational Artin induction from integral Brauer induction. It invalidates the stated cyclic integral reduction, while leaving appropriate forms of these classical induction theorems available for a revised argument.

Theorem 3.3 (Prime-tail obstruction)**.** *For every real $\sigma \leq 1$, the positive series $\sum_p p^{-\sigma}$ diverges. Thus $\sum_{p>N} p^{-\sigma}$ is not a vanishing absolutely convergent tail in the critical strip.*

Proof. It suffices to show $\sum_p 1/p = \infty$, since $p^{-\sigma} \geq p^{-1}$. Suppose the reciprocal-prime series converged. For $0 \leq x \leq 1/2$, $-\log(1 - x) \leq 2x$, for example by integrating $(1 - t)^{-1} \leq 2$ from zero to x. Hence the finite Euler products $\prod_{p \leq X}(1 - 1/p)^{-1}$ would be bounded uniformly in X. Expanding each as a finite product of absolutely convergent geometric series gives the sum of $1/n$ over all positive integers with prime factors at most X. In particular it dominates $\sum_{n \leq \lfloor X \rfloor} 1/n$, which is unbounded. Contradiction. □

Any repair below $\operatorname{Re} s = 1$ must introduce proved cancellation or a regularization, with quantitative control; absolute prime-tail bounds cannot suffice.

4 Determinants and lost arithmetic information

For $T = \operatorname{diag}(\chi(p)p^{-s})_{p \leq N}$ one has exactly

$$\det(I - T) = \prod_{p \leq N} (1 - \chi(p)p^{-s}) = L_N(s, \chi)^{-1},$$

not L_N. On an absolute-convergence region,

$$\log\det(I-T) = -\sum_{k\geq 1}\frac{\operatorname{tr}(T^k)}{k}.$$

The reciprocal orientation matters before any analytic continuation. Moreover Hermitian-part extraction does not preserve this determinant:

$$T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \frac{T+T^*}{2} = 0, \quad \det(I-T) = 2 \neq 1 = \det(I).$$

An identity for the transfer determinant therefore does not automatically identify the centered characteristic polynomial of a symmetrized operator.

Proposition 4.1 (A direct obstruction to the supplied diagonal diagnostic)**.** *For the $q = 1$ model evaluated at $s = 1/2$, the algorithm that sets $\lambda_p = -\arg(p^{-1/2})$ gives $H_N = 0$ and $F_N(s) = (s-1/2)^{d_N}$. These functions cannot converge locally uniformly to $\Lambda(s,1)$ or $\xi(s)$ on a neighborhood of $1/2$.*

Proof. Every $p^{-1/2}$ is positive, so its argument is zero and $F_N(1/2) = 0$. On the other hand the alternating Dirichlet series $\eta(1/2) = \sum_{n\geq 1}(-1)^{n-1}n^{-1/2}$ is positive: grouping into adjacent pairs gives positive terms and its sum is at least $1 - 1/\sqrt{2} > 0$. The continuation identity $\eta(s) = (1-2^{1-s})\zeta(s)$ gives $\zeta(1/2) < 0$. The gamma factor at $1/4$ is positive and finite, so $\Lambda(1/2,1) \neq 0$, and its pole-removal factor at $1/2$ is nonzero. Even pointwise convergence at that point is impossible. □

Proposition 4.2 (Holomorphic Hermitian rigidity)**.** *Let $U \subset \mathbb{C}$ be connected and open, with a fixed inner product on $\mathbb{C}^d$. If $H : U \to M_d(\mathbb{C})$ has holomorphic entries and $H(s)^* = H(s)$ for every complex $s \in U$, then H is constant.*

Proof. In fixed orthonormal coordinates, for each vector v, the holomorphic scalar $v^*H(s)v$ is real-valued, so it is constant by the open mapping theorem. Polarization recovers all matrix entries from these quadratic forms, hence every entry is constant. Equivalently, $H_{ij} + H_{ji}$ and $i(H_{ij} - H_{ji})$ are real-valued holomorphic functions whose constancy determines H_{ij}. □

Thus a nonconstant arithmetic transfer $T(s)$ cannot retain entrywise holomorphy after pointwise Hermitian-part extraction: a holomorphic $(T+T^*)/2$ would be constant, and the remaining holomorphic skew-Hermitian part would be constant as well. This excludes that particular shortcut, not fixed self-adjoint operators. The spectral criterion uses H_N independent of the spectral variable s. Requiring self-adjointness only on a real parameter line is also consistent with a nonconstant holomorphic extension, but needs its own zero-location theorem.

Proposition 4.3 (A positive generalized spectral pencil)**.** *Let $A = A^*$ and $B = B^* > 0$ be fixed $d \times d$ matrices. Every zero of $\det(A - zB)$ is real. Consequently every zero of $\det(A + i(s-1/2)B)$ lies on $\operatorname{Re} s = 1/2$.*

Proof. Singularity supplies $v \neq 0$ with $Av = zBv$. Hence $z = (v^*Av)/(v^*Bv)$, with real numerator and strictly positive real denominator. Thus z is real. Substitute $z = -i(s-1/2)$. □

This supplies a valid finite spectral mechanism, but not its arithmetic construction. To apply it one must construct A_N, B_N, certify $B_N > 0$, and prove the required completed-L determinant approximation independently. Hermitian coefficients alone are insufficient: for

$$A = \begin{pmatrix} 0 & \varepsilon \\ \varepsilon & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \det(A - zB) = -(z^2 + \varepsilon^2),$$

$0 < \varepsilon < 1/2$ gives transported off-line zeros $s = 1/2 \pm \varepsilon$ inside the critical strip.

5 A necessary change of approximation class

Theorem 5.1 (Zero-free Euler truncations cannot approximate a zero)**.** *Let U be a connected open subset of $\{\operatorname{Re} s > 0\}$, let g_N be holomorphic and nowhere zero on U, and let P_N be arbitrary finite sets of rational primes. Put*

$$D_N(s) = \prod_{p \in P_N} (1 - \chi(p) e^{-s \log p}).$$

Each of $g_N D_N$ and g_N / D_N is holomorphic and nowhere zero on U. If either sequence converges locally uniformly to a holomorphic f which is not identically zero, then f has no zero in U.

Proof. For $\operatorname{Re} s > 0$, $|\chi(p) e^{-s \log p}| \le p^{-\operatorname{Re} s} < 1$, so every local factor is nonzero. Hence the finite product, its reciprocal, and their normalized versions have the asserted property. If f had a zero, choose a closed disk inside U with a positive number of zeros of f in its interior and none on its boundary. Local uniform convergence and Rouché's theorem would force the zero-free approximants to have that positive number of zeros, a contradiction. □

More quantitatively, on any such disk D, every zero-free holomorphic approximant G satisfies

$$\max_{\partial D} |G - f| \;\ge\; \min_{\partial D} |f| > 0.$$

Otherwise Rouché yields the same contradiction. Thus improved estimates cannot repair this approximation class near a nontrivial zero. The construction must change so that finite approximants can have zeros in the strip; a nonvanishing gamma or exponential normalization cannot do this.

6 A convergent arithmetic replacement

For a nonprincipal character modulo q, put $A(x) = \sum_{n \le x} \chi(n)$ and $B = \sup_x |A(x)| \le q$. Complete periods sum to zero, so B is finite. Partial summation gives, for $\sigma = \operatorname{Re} s > 0$,

$$L(s, \chi) - \sum_{n \le N} \chi(n) n^{-s} = -A(N) N^{-s} + s \int_N^\infty A(x) x^{-s-1} \, dx,$$

and therefore

$$\left| L(s, \chi) - \sum_{n \le N} \chi(n) n^{-s} \right| \le B \left(1 + \frac{|s|}{\sigma} \right) N^{-\sigma}.$$

The endpoint tends to zero and the integral is absolutely convergent. The limit agrees with the Euler definition on $\operatorname{Re} s > 1$, hence with its analytic continuation. Multiplication by the correct gamma/conductor factor gives locally uniform approximation on $\operatorname{Re} s > 0$ for either parity.

For zeta, reproduce the pole before removing it. Set

$$Z_N(s) = \sum_{n \le N} n^{-s} + \frac{N^{1-s}}{s-1}.$$

Euler summation gives, for $\sigma > 0$ and $s \neq 1$,

$$\zeta(s) - Z_N(s) = -s \int_N^\infty \{x\} x^{-s-1}\, dx, \qquad |\zeta(s) - Z_N(s)| \le \frac{|s|}{\sigma} N^{-\sigma}.$$

The integral continues holomorphically to $\sigma > 0$. Consequently

$$\xi_N(s) = \frac{s}{2}\pi^{-s/2}\Gamma(s/2)\left((s-1)\sum_{n \le N} n^{-s} + N^{1-s}\right)$$

is holomorphic on that half-plane, including $s = 1$, and converges locally uniformly to ξ; in fact $\xi_N(1) = 1/2$ for every N. These constructions use known arithmetic coefficients, not unknown zeros. They avoid Theorem 5.1 by allowing additive cancellation and unrestricted finite-level zeros. They establish convergence, but provide no self-adjoint realization or critical-line theorem for those zeros.

7 A certified test of the arithmetic pencil candidate

Let χ be the primitive nonprincipal character modulo 3. Its fifth Dirichlet partial sum is the entire function

$$f(s) = S_5(s, \chi) = 1 - 2^{-s} + 4^{-s} - 5^{-s}.$$

Theorem 7.1 (Certified zero of a finite partial sum)**.** *Put*

$$c = (337261 + 11027504i)/10^6, \qquad r = 10^{-4}.$$

The disk $|s - c| < r$ contains exactly one zero of f, counted with multiplicity, and lies inside $0 < \operatorname{Re} s < 1/2$.

Proof. The exact rational procedure detailed in the extended edition establishes

$$|f(c)| < 10^{-6}, \qquad |f'(c)| > 1, \qquad \sup_{|s-c| \le r} |f''(s)| \le 9.$$

It uses 60 terms of the logarithm's positive atanh series and terms 0 through 120 of the exponential Taylor series, with explicit remainders; no floating-point value is used in these inequalities. On the boundary, Taylor's formula gives

$$|f(s) - f'(c)(s-c)| < 10^{-6} + \tfrac{9}{2}r^2 < r < |f'(c)(s-c)|.$$

Rouché gives exactly one zero, which is therefore simple. The real part throughout the closed disk belongs to $[337161, 337361]/10^6 \subset (0, 1/2)$. □

The completed fifth partial sum has the same zero because its conductor/gamma factor is holomorphic and nonzero on this disk. It cannot equal a nowhere-zero holomorphic factor times the determinant of a fixed positive Hermitian pencil there. This rejects that exact finite-level identification. It does not reject convergence of the partial sums, modified approximants, later truncations, subsequences, or GRH itself: f is a finite partial sum, not the full L-function.

8 Jensen approximants and a coefficient-defined metric

This alternative currently concerns the Riemann target only. Define

$$\gamma_j = \frac{j!\,\xi^{(2j)}(1/2)}{(2j)!}, \qquad F(w) = \sum_{j\ge 0} \frac{\gamma_j}{j!} w^j, \qquad \Xi(t) = \xi(1/2 + it) = F(-t^2).$$

The classical positive Fourier kernel of Ξ gives $\gamma_j > 0$ [9]; evenness and entirety of ξ make F entire. Set, for $d \ge 1$,

$$J_d(x) = \sum_{j=0}^{d} \binom{d}{j} \gamma_j x^j, \qquad Q_d(w) = J_d(w/d), \qquad P_d(t) = Q_d(-t^2).$$

Proposition 8.1 (Explicit convergence without a zero-location premise)**.** *For $R, T \ge 0$,*

$$\sup_{|w|\le R} |Q_d(w) - F(w)| \le \frac{R^2 F''(R)}{2d}, \qquad \sup_{|t|\le T} |P_d(t) - \Xi(t)| \le \frac{T^4 F''(T^2)}{2d}.$$

Proof. The coefficient multiplier $w_{d,j} = j!\binom{d}{j}/d^j$ for $j \le d$, and 0 for $j > d$, satisfies $0 \le w_{d,j} \le 1$ and $1 - w_{d,j} \le j(j-1)/(2d)$. For $j \le d$ use $1 - \prod_{k<j}(1 - k/d) \le \sum_{k<j} k/d$; for $j > d$ use $j(j-1) \ge 2d$. Summing the absolutely convergent series with nonnegative coefficients proves the bounds. □

Let C_d be the multiplication-by-x companion of the monic J_d/γ_d, and $H_d = (m_{i+j})_{0\le i,j<d}$, where $m_k = \operatorname{tr}(C_d^k)$ and $m_0 = d$. Newton identities compute H_d from the coefficients, and $H_d C_d = C_d^T H_d$. The Hermite–Sylvester criterion gives $H_d \ge 0$ exactly when J_d is real-rooted [8]; then its positive coefficients force all roots negative. If $H_d > 0$, the extended edition constructs an explicit real symmetric K_d with

$$P_d(t) = \frac{(-1)^d \gamma_d}{d^d} \det(tI - K_d).$$

Strict positivity is needed for that construction; semidefinite metrics can be singular. The classical Jensen criterion makes $H_d \ge 0$ for every d equivalent to zeta RH [7]. This identifies the missing positivity; it does not prove it. Fixed-degree eventual hyperbolicity as the shift grows does not settle this unshifted growing-degree family.

Proposition 8.2 (Positive smooth kernels do not supply the metric)**.** *Positivity of an even smooth Fourier density, including Gaussian decay, does not imply positivity of its Jensen Newton-sum metric.*

Proof. Convolve $\frac{9}{10}\delta_0 + \frac{1}{20}(\delta_{-1} + \delta_1)$ with a centered Gaussian of variance $v > 0$. Its Fourier transform is $e^{-vt^2/2}(9/10 + \cos t/10)$ and its density is strictly positive, even and Schwartz. Writing its associated $F(w) = \sum \gamma_j w^j/j!$ gives $\gamma_0 = 1$, $\gamma_1 = (v + 1/10)/2$ and $\gamma_2 = (3v^2 + 3v/5 + 1/10)/12$. Thus $\gamma_1^2 - \gamma_0\gamma_2 = -7/1200 < 0$; the degree-two Newton-sum metric has negative determinant. These are model coefficients, not the arithmetic ξ coefficients. □

9 Extending an Euler correction and locating an additional factor

We now test one missing analytic input of the September argument in [3]. All deductions in this section use its displayed scalar local identity; its preceding Gauss/reciprocity derivation and main mean-square theorem are not certified here.

Proposition 9.1 (Extension of the explicit local product)**.** *The displayed correction product* $\mathcal{H}_{\eta,u}(x,w,z)$ *converges normally on a neighborhood of*

$$\operatorname{Re} x \geq 3/4, \qquad \operatorname{Re} w \geq 19/20, \qquad \operatorname{Re} z \geq 33/200,$$

and is $O_\epsilon(q_u^\epsilon)$ *there. On the displayed closed region, for* $Q = Np$*, its local defects are at most* $1024Q^{-149/100}$ *when* $p \nmid u$ *and* $32Q^{-7/10}$ *when* $p \mid u$*. The latter is an upper bound, not a nonvanishing statement.*

Proof. Substitution in the explicit local formulas gives the exponent and denominator bounds detailed in the extended edition. All five positive row valuations are included. The ideal count sums the first bound; the divisor-product estimate handles the finitely many primes of u. The bounds retain a positive margin on a larger open region. □

For the principal row put $H_\eta(s) = \mathcal{H}_{\eta,1}(s,1,1/6)$. At a prime set

$$v = Q^{-1}, \quad d = \eta(p)Q^{-s}, \quad \theta(p) = \alpha(p)^{-6}\eta(p)^6, \quad r = \theta(p)Q^{3-6s},$$

where $\alpha(p) = p/|p|$ for a generator. Sixth powers remove unit ambiguity. Exact simplification gives

$$H_p = \frac{(1-v^2)[1-vr-(1-v)d]}{(1-r)(1-d)}, \qquad A_p = (1-r)H_p.$$

For $\operatorname{Re} s > 1/2$, the numerator is nonzero because $|vr + (1-v)d| < v + (1-v) = 1$. Moreover

$$A_p - 1 = -v^2 + \frac{(1-v^2)v(d-r)}{1-d}.$$

Thus $A = \prod_{p \notin S} A_p$ is holomorphic and nonzero on $\operatorname{Re} s > 1/2$, while H_η itself converges normally and is nonzero on $\operatorname{Re} s > 2/3$. On the latter region the exact factorization is

$$H_\eta(s) = A(s) L_F^S(6s-3, \theta).$$

The character θ has nontrivial angular factor α^{-6}; it has infinite order and is outside the source's finite-order target family. The correction can be made uniformly close to 1 on each fixed $\operatorname{Re} s \geq \sigma_0 > 2/3$ by a sufficiently large finite deletion. This supplies the proposed tiny perturbation's local Euler input; the compensated contour errors and mean squares remain unproved.

Proposition 9.2 (Limitation of the unchanged correction)**.** *Assume the scalar identity and the classical Hecke completion properties listed in the extended edition, including order one and conjugation. The analytic continuation of this unchanged* H_η *has a zero with real part at least* $7/12$*. It therefore cannot be nonzero throughout* $\operatorname{Re} s > \sigma_0$ *for any* $\sigma_0 < 7/12$*.*

Proof. The entire order-one completion of $L(s,\theta)$ has infinitely many zeros: otherwise Hadamard factorization would contradict its $\sigma \log \sigma + O(\sigma)$ logarithmic growth on the positive axis. Euler nonvanishing, the functional equation and conjugation give a zero ρ with $1/2 \leq \operatorname{Re}\rho \leq 1$. Neither gamma nor deleted Euler factors vanish there. Hence the displayed continuation vanishes at $(\rho+3)/6$, where A is nonzero and $\operatorname{Re} s \geq 7/12$. □

The Hecke analytic inputs are classical external dependencies, not Lean theorems here [10, 11]. No GRH or Artin holomorphy premise is used. This limits the specified probe and nonvanishing criterion; it does not refute GRH, the claimed 7/8 result, or different constructions.

10 The compensated principal residue at perturbed parameters

We retain the source's scalar local identity, exact high representation, Mellin integrated/trace tail contracts, and polynomial reciprocal control on $\operatorname{Re} s = \beta_* + e$. The following verifies a principal interface, not the remaining energy or row-count estimates. On the enlarged Euler region, the four local multiplier errors have powers

$$Q^{-x_r}, \quad Q^{-6z_r}, \quad Q^{4-5x_r-6z_r}, \quad Q^{1-w_r-6z_r}.$$

Their common saving is at least $37/50$. The local lower bound for H_p therefore gives $\mathcal{B}_p = -1 + O(Q^{-37/50})$, uniformly in all heights. On fixed real boxes the full principal correction is $O(Z^{\ell \operatorname{Re} z})$, with height degree zero.

For a nonempty fixed slot system with positive lengths ℓ_i, set

$$S_i = \sum_{p \in \mathcal{P}_i(Z)} W_i(Q/P_i) Q^{-5/6}, \qquad A_T = (-1)^K Z^{-\ell/6} \prod_i S_i, \qquad \mu = \tfrac{7}{10} \min_i \ell_i > 0.$$

Fixed-ray prime counting gives $S_i > 0$ eventually and $A_T \sim c_T (\log Z)^{-K}$ with $c_T \neq 0$, of sign $(-1)^K$ [12]. No uniformity in a moving ray conductor is used. Nonnegative weights give the exact residue factorization

$$\mathfrak{H}_{\eta,1,Z}(s, 1, 1/6) = H_\eta(s) Z^{\ell/6} A_T(Z)(1 + \mathcal{R}_{\eta,Z}(s)), \qquad \mathcal{R}_{\eta,Z} = O(Z^{-\mu}).$$

Indeed the ith relative error is the negative weighted average of $\mathcal{B}_p + 1$, of size $O(P_i^{-37/50})$; a fixed finite product preserves that saving. Also $|A_T|^{-1} \ll_\epsilon Z^\epsilon$.

With the explicitly inherited contracts above, the source's principal contour comparison now extends to $3/4 \le \sigma_0 < \beta_*$. Its three remainders, relative to $C(\beta_*)$, have exponents

$$(1 + h)e - l_y/20 + \epsilon, \qquad e - h/600 + \epsilon, \qquad e - \mu + \epsilon.$$

The extended edition derives these by crossing only $w = 1$ and $z = 1/6$. All three are negative for sufficiently small pretarget e, ϵ. The perturbed geometry at $t = 10^{-5}$ satisfies this interface, with $C(s) = s - 11/16$ and target $349999/400000$. This is not a new zero-free theorem: other low, nonprincipal contour and row-count inputs remain unverified.

There is a separate pointwise limitation. At the principal residue, putting $N = 1 - vr - (1 - v)d$ gives

$$\mathcal{B}_p + 1 = \frac{(1 - v)[r/d - vr - (1 - v)d]}{N}.$$

For fixed $1/2 < \sigma = \operatorname{Re} s \le 3/5$ its norm is asymptotic to $Q^{3-5\sigma}$, uniformly in phases. In particular, for $\eta = 1$ and an inert rational prime, $\alpha^{-6} = 1$ and real $s = 3/5$ give $r = d$, so $\mathcal{B}_p = -1/Q$ exactly. The error around -1 tends to 1. Such primes can occur in the fixed identity ray. This excludes the pointwise power-saving argument there, without excluding cancellation after prime averaging or a different normalization.

11 Consequences for a revised research article

Theorems 3.1, 3.2, and 3.3 require replacements for the associated proof steps, not additional exposition. A credible next theorem would construct operators independently of unknown zeros and establish their exact arithmetic determinant, including ramified, archimedean, and conductor terms.

It must then prove locally uniform convergence or the local boundedness in Proposition 2.3. Only then does Theorem 2.1 prove the desired zero location.

Weil/Li positivity, adelic trace methods, and canonical-product approximations are potential alternative interfaces. They move the research obligation to a specific positivity, trace, or approximation theorem; they do not discharge it by analogy. Lagarias [4] and Connes [5] provide relevant primary-source starting points. No novelty or complete proof is claimed for these classical conditional principles.

A Worked geodesic calculations and possible replacements

For trace $t = 3$, the expanding eigenvalue is $(3 + \sqrt{5})/2$ and

$$N = \frac{7 + 3\sqrt{5}}{2}, \qquad \ell = 2 \log \frac{3 + \sqrt{5}}{2}.$$

This is a legitimate hyperbolic length, but not $\log p$. For $p = 2$, the proposed length $\log 2$ would require $t = 3/\sqrt{2}$, equivalently $2t^2 = 9$, which is impossible even modulo two. The Lean companion verifies this last exact integer obstruction. For arbitrary p^k, the integer argument in Theorem 3.1 is stronger than a numerical search over traces.

A Hecke operator attached to determinant-p double cosets is a different object from a determinant-one deck transformation. A replacement must specify the ambient space, coefficient system, trace class or regularized trace, and the precise arithmetic distribution. Reusing the same symbol for these objects does not construct an identification. An adelic framework may encode prime norms intrinsically; its trace and positivity obligations must then be proved in that framework.

B Character examples and induction bookkeeping

The nonprincipal character modulo three has values $\chi(1) = 1$, $\chi(2) = -1$, $\chi(0) = 0$ and is odd. For scalar weight-zero automorphy under $\Gamma_0(3)$,

$$f(\gamma z) = \chi(d) f(z),$$

the element $-I$ fixes z and has $\chi(-1) = -1$. Hence every such function would satisfy $f(z) = -f(z)$ and vanish. The proposed scalar bundle on the projective quotient is therefore unavailable as written. One can work on a different group or choose a compatible nonzero weight; either change requires rebuilding the operator and trace identity.

Modulo five, the quadratic character has $\chi(2) = -1$. Consequently the one-prime diagonal entry $\chi(2)2^{-1/2}$ is negative. Its 1×1 minor is already negative, so an unrestricted assertion of total nonnegativity of these transfer matrices is false. A positive state constructed as $TT^*/\operatorname{tr}(TT^*)$ discards phase information in this diagonal model: the nonzero character phases disappear from TT^* at real s, although the ramification zero mask remains. Thus positivity of that state cannot by itself recover character-sensitive Euler factors.

For $C_2 \times C_2$, the three order-two subgroups each admit two linear characters. All six induced characters have degree two; induction from the trivial subgroup gives degree four. Negative integer coefficients do not fix the parity obstruction. Rational coefficients can change parity, which explains why rational and integral induction cannot be interchanged. For a monomial representation $\operatorname{Ind}_H^G \theta$, the appropriate Artin formalism yields a Hecke L-function over the fixed field of H; it does not automatically reduce to a product of Dirichlet functions over $\mathbb{Q}$. Every extension must retain the base field and infinity type.

C Local algebra and topology checks

For the upper-Iwahori order

$$\mathcal{O} = \begin{pmatrix} R & R \\ \pi R & R \end{pmatrix}, \quad S = \begin{pmatrix} 0 & \pi \\ 1 & 0 \end{pmatrix},$$

$E_{12} \in \mathcal{O}$, whereas $SE_{12}S^{-1} = \pi^{-1}E_{21} \notin \mathcal{O}$. The displayed shift orientation therefore does not normalize this order. Changing an orientation may repair this isolated calculation, but all subsequent products and gluing maps must be rechecked with that convention.

For $R = \mathbb{C}[[t]]$ and $\lambda \in \mathbb{C}^\times$, there is no unital $\mathbb{C}$-algebra homomorphism sending t to λI in a nonzero matrix algebra: $t - \lambda$ is a unit in R and would be sent to zero. Thus substituting a unit complex phase for a formal uniformizer is not a representation of this allowed complete DVR. An algebra of Laurent polynomials can admit unitary phase evaluation, but it is a different algebra and needs its own construction.

A dessin on a sphere with two vertices and three edges has three faces, by Euler's formula $V - E + F = 2$, not two. Repeating a vertex along a face boundary does not create a new primitive vertex idempotent. In particular, a three-vertex path algebra with arrows between distinct vertices cannot have the claimed surjection to a six-vertex radical-square-zero cycle algebra: compose with the target's semisimple quotient k^6. Arrows map to zero, since the orthogonal source idempotents force their endpoint supports to be disjoint in the commutative target. The image is generated by at most three orthogonal idempotents, so it cannot be all of k^6. A valid replacement may require a cover or unfolded category rather than a quotient of the stated algebra.

D Why weak convergence and boundary analogies are insufficient

Let P_n be projection onto the nth basis vector of ℓ^2. For every $x \in \ell^2$, $\|P_n x\| = |x_n| \to 0$, so $P_n \to 0$ strongly, but $\|P_n\|_1 = 1$ for every n. Moreover $\det(I - P_n) = 0$ while $\det(I) = 1$. Thus even strong operator convergence does not imply determinant convergence. Trace-norm control is a natural sufficient condition, and must be established for the actual operator family, locally uniformly in s, before invoking Fredholm determinant continuity.

For a Phragmén–Lindelöf argument, one needs uniform control of approximants as the approximation level changes. Polynomial growth of each individual polynomial has constants and degrees depending on N; it does not supply a common bound. Dividing by $(1 + |s|^2)^M$ produces a nonholomorphic expression, to which the maximum modulus principle does not apply. Dividing by $(1 + s^2)^M$ instead introduces poles at $s = \pm i$; a denominator must be holomorphic and nonvanishing on the actual domain. Even after selecting such a denominator, the necessary uniform boundary and horizontal-side estimates are substantive mathematical work.

Likewise, a bound on a density of geodesics is not a bound on a weighted analytic remainder. The full test-function dependence, prime powers, scattering, elliptic and parabolic contributions must be retained. On a cusped finite-area surface the infimum of the injectivity radius is zero. It cannot provide the positive global mesh bound claimed in the source completion argument.

E Spectral approximation can conceal the original hypothesis

Proposition E.1 (Canonical-product converse for the Riemann target)**.** *If RH holds, the entire order-one function ξ admits locally uniform approximations of the form in Theorem 2.1 with diagonal self-adjoint H_N.*

Proof. The Hadamard factorization of ξ, which has no zero at zero, is

$$\xi(s) = e^{a+bs} \prod_{\rho} (1 - s/\rho) e^{s/\rho},$$

with an enumeration for which the canonical product converges locally uniformly. Under RH every $\rho = 1/2 + i\gamma$ has real γ. Each finite product is an exponential of an affine function times a constant times $\prod_{j \le N}(s - \rho_j)$; absorb the nonzero constant into the exponential. Take $H_N = \operatorname{diag}(\gamma_1, \ldots, \gamma_N)$. □

Together with Theorem 2.1, this shows why existence of the approximants, without an independent arithmetic construction, may merely restate RH. It is not progress to import the unknown zero locations into the input spectrum and then prove that the constructed spectrum is real. This proposition is a classical consequence, not a novel proof of RH.

F Hermitian dilation and the spectral parameter

A common attempt to preserve arithmetic input while obtaining self-adjointness is the dilation, for square $T(s) \in M_d(\mathbb{C})$,

$$K(s) = \begin{pmatrix} 0 & T(s) \\ T(s)^* & 0 \end{pmatrix}.$$

It is Hermitian pointwise. Singular-value decomposition shows that its spectrum is $\{\pm\sigma_j(T(s)) : 1 \le j \le d\}$, with multiplicities. Consequently

$$\det(\lambda I - K(s)) = \prod_{j=1}^{d} (\lambda^2 - \sigma_j(T(s))^2), \qquad \det(-K(s)) = (-1)^d |\det T(s)|^2.$$

These formulas retain singular values and erase determinant phase. For a nonconstant holomorphic scalar f, $|f|^2$ cannot be holomorphic: a real-valued holomorphic function is constant, and constant modulus then forces f constant. Thus the zero-matching scalar $|f|^2$ is not a holomorphic replacement for f. Matrix nonholomorphy alone does not imply that every scalar expression formed from it is nonholomorphic; each proposed determinant identity must be examined separately. For example, $T(s) = \operatorname{diag}(e^s, e^{-s})$ is nonconstant but its dilation determinant is constantly one.

For the prime-diagonal model, the dilation eigenvalues are exactly $\pm|\chi(p)|p^{-\operatorname{Re} s}$. Both $\Im s$ and the nonzero character phases disappear. This cannot encode the proposed arithmetic zero ordinates. The source's parameter mismatch is visible where an s-dependent transfer is converted to a Hermitian/modular Hamiltonian and then used as though it had fixed eigenvalues in a characteristic polynomial [1], source lines 11465–11484 and 12200–12215; see also lines 13636–13643 for the explicit skew-Hermitian-part extraction. Its numerical code fixes a particular evaluation point instead, so the rigidity proposition does not itself refute that distinct operation. The earlier diagonal diagnostic obstruction addresses it directly.

A permitted architecture is a fixed self-adjoint H_N with spectral variable entering through $sI - (1/2 + iH_N)$. Alternatively one may use a holomorphic pencil that is Hermitian only on its real parameter axis, but this condition does not force real spectral roots. Already the scalar polynomial $h(t) = t^2 + 1$ is real for real t, whereas $z - h(z) = 0$ has the nonreal solutions $(1 \pm i\sqrt{3})/2$. An infinite-dimensional operator model would additionally need explicit domains, closure, adjoint and determinant regularization. No finite-matrix rigidity statement here is asserted to settle variable-domain unbounded operator constructions.

G Positive pencils, resolvent control, and a Herglotz test

The positive generalized pencil is precisely a fixed self-adjoint construction in different coordinates. Put $H = B^{-1/2}AB^{-1/2} = H^*$. Then

$$A - zB = B^{1/2}(H - zI)B^{1/2},$$

and therefore

$$\det(A + i(s - 1/2)B) = i^d \det(B) \det\big(sI - (\tfrac{1}{2}I + iH)\big).$$

The positive metric B must be supplied by the arithmetic construction; its existence is not inferred from a desire for real roots.

There is also a quantitative test. If $B \geq bI$ with $b > 0$ and z is nonreal, then for every v,

$$\|(A - zB)v\| \, \|v\| \geq |v^*(A - zB)v| \geq |\Im z| v^*Bv \geq b|\Im z| \, \|v\|^2.$$

For $v \neq 0$, divide by $\|v\|$. In finite dimension this proves invertibility and the operator-norm bound

$$\|(A - zB)^{-1}\| \leq \frac{1}{b|\Im z|} = \frac{1}{b|\operatorname{Re} s - 1/2|}.$$

Uniform approximation arguments must track dimension, normalization and b_N; a bound for each individual pencil does not by itself provide uniform determinant convergence.

A nonlinear extension uses a matrix-valued Herglotz condition: if $M(z)$ is holomorphic upstairs and

$$\Im M(z) := \frac{M(z) - M(z)^*}{2i} > 0,$$

then $M(z)v = 0$ would contradict $v^*\Im M(z)v > 0$. Thus $\det M(z) \neq 0$ there. A reflected lower-half-plane extension $M(\bar{z}) = M(z)^*$ is likewise invertible. A quantitative lower bound $\Im M(z) \geq \eta(z)I > 0$ gives $\|M(z)^{-1}\| \leq 1/\eta(z)$ by the same argument. For the positive linear pencil the Herglotz sign is $M(z) = zB - A$, not $A - zB$. Semidefinite positivity alone allows $M(z) = \operatorname{diag}(z, 0)$, which is always singular. Without reflection, $M(z) = z + i$ is strictly Herglotz upstairs but has a lower-half-plane zero. Possible real-axis poles and the arithmetic determinant identity remain separate obligations.

For a candidate normalized identity written in the z-coordinate, $F_N(z)/g_N(z) = \det(A_N - zB_N)$ with a prescribed nowhere-zero g_N, a necessary diagnostic, for $d_N > 0$, is

$$-\partial_z \log(F_N/g_N) = \operatorname{tr}\big((A_N - zB_N)^{-1}B_N\big) = \sum_{j=1}^{d_N} \frac{1}{\lambda_{j,N} - z}.$$

Its imaginary part is strictly positive for $\Im z > 0$. A certified violation rejects that particular determinant identity; finitely many successful samples cannot prove it. One must not choose g_N using unknown zero locations or assume this Herglotz positivity as a substitute for establishing the arithmetic bridge.

H Quantitative arithmetic approximation and prime regularization

For the nonprincipal-character approximants, a compact set $K \subset \{\operatorname{Re} s > 0\}$ has $\delta = \min_K \operatorname{Re} s > 0$, $M = \max_K |s|$, and $G = \max_K |(q/\pi)^{(s+a)/2}\Gamma((s + a)/2)| < \infty$. The completed approximation error is at most $GB(1 + M/\delta)N^{-\delta}$ uniformly on K. This is a fixed-character compact bound, not a

uniform-in-conductor GRH estimate. To verify the Abel formula's sign, first sum between integers $N < T$:

$$\sum_{N<n\le T} \chi(n)n^{-s} = A(T)T^{-s} - A(N)N^{-s} + s\int_N^T A(x)x^{-s-1}\,dx.$$

Then $A(T)T^{-s} \to 0$. Bounding $|A|$ by B gives exactly the main-text estimate. For zeta, $\lfloor x\rfloor = x-\{x\}$ in the same summation formula yields

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \{x\}x^{-s-1}\,dx.$$

The finite sum formula gives the stated remainder after subtracting Z_N. The integral and its derivatives converge uniformly on compact subsets of the positive half-plane, since logarithmic powers are integrable against $x^{-1-\delta}$. This justifies holomorphy and continuation. Multiplying the remainder by $s(s-1)\pi^{-s/2}\Gamma(s/2)/2$ gives the compact-uniform ξ_N error, including at $s = 1$. In contrast, multiplying a finite Euler product by $s(s-1)$ would force a zero at 1 for every truncation, although $\xi(1) = 1/2$.

A second exact interface isolates what a prime-based repair would need. For a nonprincipal primitive χ, define on $\operatorname{Re} s > 1$

$$P_\chi(s) = \sum_p \chi(p)p^{-s}, \qquad R_\chi(s) = \sum_p \sum_{k\ge 2} \frac{\chi(p)^k}{kp^{ks}}.$$

On each compact subset of $\operatorname{Re} s > 1/2$, the second series converges absolutely and uniformly. Indeed, for $\operatorname{Re} s \ge \delta > 1/2$,

$$\sum_{k\ge 2} \frac{|\chi(p)|^k}{kp^{k\delta}} \le \frac{p^{-2\delta}}{1-2^{-\delta}},$$

and the prime sum is dominated by the convergent integer series $\sum_{n\ge 2} n^{-2\delta}$. Thus R_χ is holomorphic there. On $\operatorname{Re} s > 1$, $L(s,\chi) = \exp(P_\chi(s) + R_\chi(s))$.

Proposition H.1 (The missing prime-linear term)**.** *For a nonprincipal primitive character, P_χ has a single-valued holomorphic continuation to* $\operatorname{Re} s > 1/2$ *if and only if $L(s,\chi)$ is zero-free there.*

Proof. If such a continuation exists, its exponential with R_χ agrees with L on $\operatorname{Re} s > 1$ and hence on the connected half-plane by the identity theorem. The exponential is nonzero. Conversely, if L is nonzero, the simply connected half-plane admits a holomorphic logarithm of L. Choose its additive constant to agree with the absolutely convergent Euler logarithm on $\operatorname{Re} s > 1$. Subtract R_χ to obtain the required continuation of P_χ. □

This proposition isolates a precise analytic obligation; it does not prove it. Regularizing away the terms $k \ge 2$ supplies no proof that the missing prime-linear function extends. The principal zeta case must first remove the pole, for example by using $(s-1)\zeta(s)$; an unqualified logarithm of ζ cannot be holomorphic through 1. The functional equation relates a fixed character's zeros across the critical line after complex conjugation, so a full right-half-plane nonvanishing theorem would give the desired Dirichlet GRH conclusion.

I Exact arithmetic behind the finite-level zero certificate

This section gives all parameters needed to reproduce Theorem 7.1. Write $C = |\operatorname{Re} c| + |\Im c| < 12$. Use the following rational logarithm approximations ℓ_n and error bounds δ_n:

n	ℓ_n	δ_n
2	$6931471805599453/10^{16}$	10^{-15}
4	$13862943611198906/10^{16}$	$2 \cdot 10^{-15}$
5	$16094379124341004/10^{16}$	10^{-15}

For $y = (n-1)/(n+1)$, set

$$L_n = 2\sum_{k=0}^{59} \frac{y^{2k+1}}{2k+1}, \qquad T_n = \frac{2y^{121}}{121(1-y^2)}.$$

The positive series for $\log n$ gives $L_n \le \log n \le L_n + T_n$. Exact rational comparisons give $\ell_n - \delta_n \le L_n$ and $L_n + T_n \le \ell_n + \delta_n$. They also give $0 < \log 2 < 1$ and $0 < \log 4, \log 5 < 2$.

Define Gaussian rationals

$$E_n = \sum_{k=0}^{120} \frac{(-c\ell_n)^k}{k!}, \qquad \tau = \frac{3^{24}24^{121}}{121!} < 10^{-18}, \qquad \eta_n = \tau + 2C\delta_n.$$

Since $|c\ell_n| \le C\ell_n < 24$, the exponential series tail is at most $e^{24}24^{121}/121! < \tau$. Here $e < 3$ follows directly from its series. Also

$$|e^{-c\log n} - e^{-c\ell_n}| \le e^{-\operatorname{Re}(c)\ell_n}|c|\delta_n e^{|c|\delta_n} \le 2C\delta_n,$$

because $\ell_n, \operatorname{Re} c > 0$, $C\delta_n \le 1/2$, and $e^u \le (1-u)^{-1} \le 2$ for $0 \le u \le 1/2$. Thus $|n^{-c} - E_n| \le \eta_n$.

Put

$$F = 1 - E_2 + E_4 - E_5, \qquad D = \ell_2 E_2 - \ell_4 E_4 + \ell_5 E_5.$$

All entries are exact Gaussian rationals. With $\|(a,b)\|_1 = |a| + |b|$, the script checks the rational inequalities

$$\|F\|_1 + \eta_2 + \eta_4 + \eta_5 < 10^{-6}, \qquad |\operatorname{Re} D| - \sum_{n\in\{2,4,5\}} (\delta_n + \ell_n\eta_n) > 1.$$

The first bounds $|f(c)|$. For the second, note that $|n^{-c}| \le 1$ and

$$|(\log n)n^{-c} - \ell_n E_n| \le \delta_n + \ell_n\eta_n.$$

This proves $|f'(c)| > 1$. On the entire closed disk, $\operatorname{Re} s > 0$, so

$$|f''(s)| \le (\log 2)^2 + (\log 4)^2 + (\log 5)^2 < 1 + 4 + 4 = 9.$$

Integrating the second derivative along the line segment from c to s justifies the Taylor remainder bound $9|s-c|^2/2$. The exact strict boundary margin is

$$r - 10^{-6} - \tfrac{9}{2}r^2 = \frac{19791}{200000000} > 0.$$

Consequently $|f(s)|$ is greater than that margin on $|s-c| = r$. Any zero-free holomorphic positive-pencil model G on a neighborhood of this disk must have boundary error at least $\min_{|s-c|=r}|f(s)|$; otherwise another application of Rouché would contradict its zero count. This is a quantitative

rejection of a prescribed finite-level model, not a proof about the zero locations of the limiting L-function.

The certificate program `scripts/check_arithmetic_zero.py` uses only Python integer and Fraction arithmetic. A separate independent exact-arithmetic reproduction passed. Lean verifies the disk's location and the last rational margin; the logarithm/exponential error estimates and the Rouché inference remain explicit prose proofs, not claims of a complete kernel-checked zero-existence theorem.

J Jensen companions: exact normalization and the failed moment argument

Our γ_j equals one eighth of the convention in equation 1 of [7], whose generating function is $8\xi(1/2+z)$. This common positive factor changes no polynomial zeros. Entirety of F follows directly from the even Taylor series: its absolute series at $|w| = R$ is the absolute series for $\xi(1/2+z)$ at $|z| = \sqrt{R}$. The finite-product inequality in the main-text convergence proof is also verified in Lean. The analytic summation and ξ specialization remain prose arguments, using classical properties of ξ. For an explicit coefficient interface, the classical representation in [9, equations 1.8–1.11] gives

$$\Phi(u) = 2\sum_{n\geq 1}(2\pi^2 n^4 e^{9u/2} - 3\pi n^2 e^{5u/2})e^{-\pi n^2 e^{2u}}, \qquad \gamma_j = \frac{j!}{(2j)!}\int_{\mathbb{R}} u^{2j}\Phi(u)\,du.$$

The kernel is even. Each summand is positive for $u \geq 0$ because $2\pi n^2 e^{2u} > 3$, so the moments are strictly positive. Its rapid decay permits differentiation of the Fourier integral. This arithmetic kernel representation is an external classical input, not a new Lean theorem in this project.

J.1 The coefficient construction

Write $p(x) = J_d(x)/\gamma_d = x^d + \sum_{i<d} a_i x^i$. The companion C has $C_{j+1,j} = 1$ for $0 \leq j < d-1$, last column $C_{i,d-1} = -a_i$, and all other entries zero. For $1 \leq k \leq d$ Newton's recurrence is

$$m_k + \sum_{i=1}^{k-1} a_{d-i} m_{k-i} + k a_{d-k} = 0,$$

and for $k > d$ it is

$$m_k + \sum_{i=1}^{d} a_{d-i} m_{k-i} = 0.$$

Together with $m_0 = d$, this determines the entire required matrix $H = (m_{i+j})_{0\leq i,j<d}$. Multiplication in the quotient algebra $\mathbb{R}[x]/(p)$ gives its bilinear form

$$h(u,v) = \mathrm{tr}(M_{uv}), \qquad h(xu,v) = h(u,xv),$$

so $HC = C^T H$ without any zero-location assumption. Here M_{uv} means multiplication by uv in that finite quotient. When roots are real and distinct, the evaluation matrix V gives $H = V^T V > 0$. Repeated roots make V singular; the semidefinite Hermite criterion still applies, but inverse square roots do not. Nonnegative leading principal minors alone are not a PSD certificate; all principal minors, or a valid alternative test, are required.

Assume now $H > 0$. Then

$$S = H^{1/2}(-C)H^{-1/2} = S^T > 0, \qquad K = \begin{pmatrix} 0 & \sqrt{dS} \\ \sqrt{dS} & 0 \end{pmatrix} = K^T.$$

Indeed the similarity is symmetric by $HC = C^T H$, and its eigenvalues are negatives of the roots of p, all positive. The block determinant and companion characteristic polynomial give

$$\det(tI_{2d} - K) = \det(t^2 I_d - dS) = \det(t^2 I_d + dC),$$
$$J_d(-t^2/d) = \gamma_d(-1/d)^d \det(t^2 I_d + dC).$$

The identity holds also at $t = 0$, by polynomial equality. Using a companion for the scaled Q_d instead would remove the extra d inside the square root. With our stated convention it cannot be omitted. Thus the arithmetic data and approximation rate are specified; the unproved universal positivity is precisely where the RH difficulty lies. This construction supplies no general Dirichlet, Hecke, Dedekind or Artin conclusion. Their symmetry and coefficient interfaces must be established separately.

J.2 Why ordinary moment positivity fails

For $p = x^2 + bx + a$, the companion and Newton-sum metric are

$$C = \begin{pmatrix} 0 & -a \\ 1 & -b \end{pmatrix}, \qquad H = \begin{pmatrix} 2 & -b \\ -b & b^2 - 2a \end{pmatrix}.$$

Direct calculation gives $HC = (HC)^T$, $\det H = b^2 - 4a$, and

$$u^T H u = 2(u_0 - bu_1/2)^2 + \tfrac{1}{2}(b^2 - 4a)u_1^2.$$

These identities and positivity under $b^2 - 4a > 0$ are formalized in Lean. For $J_2 = \gamma_0 + 2\gamma_1 x + \gamma_2 x^2$, $\gamma_2 > 0$,

$$\det H = \frac{4(\gamma_1^2 - \gamma_0\gamma_2)}{\gamma_2^2}.$$

The unsmoothed atomic example has $\mu_0 = 1$, $\mu_2 = \mu_4 = 1/10$. Its ordinary even-moment Hankel matrix and Jensen metric are respectively

$$\begin{pmatrix} 1 & 1/10 \\ 1/10 & 1/10 \end{pmatrix} > 0, \qquad \begin{pmatrix} 2 & -12 \\ -12 & -96 \end{pmatrix}, \quad \det H = -336.$$

Gaussian smoothing gives the strictly positive densities in the main-text counterexample. The standard Gaussian moments yield $\mu_2 = v + 1/10$ and $\mu_4 = 3v^2 + 3v/5 + 1/10$. With $\gamma_n = n!\mu_{2n}/(2n)!$, the negative gap is independent of v. Lean checks that exact algebraic cancellation; the measure integral and Fourier-transform identities remain prose. Thus even smooth positive kernel information cannot replace arithmetic-specific Jensen inequalities.

The exact program `scripts/check_jensen_metric.py` verifies the Newton recurrence, companion symmetry, positive LDL pivots for the $\cosh\sqrt{w}$ model at degrees 1 through 8, and the negative and singular witnesses. Its finite coefficient-bound checks supplement the universal proof; they do not replace it. No arithmetic ξ coefficient intervals or universal Jensen positivity are certified by this program. For example the sequence $(1, 1, 1)$ gives $J_2 = (1 + x)^2$ and $H = \left(\begin{smallmatrix} 2 & -2 \\ -2 & 2 \end{smallmatrix}\right)$, which is PSD and singular. Adding an arbitrary positive diagonal to H need not preserve $HC = C^T H$.

K Euler-correction estimates and the shifted angular character

The immutable scalar input is the September manuscript's local identity, lines 3845–4206 at the pinned source in the project manifest. We verify its rational consequences, without claiming a new verification of the underlying Gauss identities. Write $Q = Np \geq 2$ and retain its notation $V = Q^{-6z}$, $R = a_p^2 Q^{4-6x-6z}$, $W = \chi_p(u)Q^{-w}$, $D = \eta(p)\overline{\chi_p(u)}Q^{-x}$. The unit phases have norm one. At $p \mid u$, $D = W = 0$ by zero extension, while the local unit phase in P_p^* is retained.

K.1 All valuation cases in the larger region

For $p \nmid u$, let $\mathcal{E} = P_p^* + D$. The $j = 0$ formula gives

$$\mathcal{E} = \frac{R((1-Q^{-1})/(1-V) + W - D) - DW(Q-1)V/(1-V)}{1-R}.$$

In the region of the main-text proposition, the exponents of $R, QDWV, DV, DW, DVW, VW$ are respectively at most

$$-\frac{149}{100}, \ -\frac{169}{100}, \ -\frac{174}{100}, \ -\frac{17}{10}, \ -\frac{269}{100}, \ -\frac{194}{100}.$$

Each of $|R|, |V|, |D|, |W|$ is below $3/4$, so the inverses of the displayed $1-$factor denominators are at most 4. Consequently $|\mathcal{E}| \leq 40Q^{-149/100}$. The exact defect identity

$$H_p - 1 = \frac{D(V + W - VW) - VW + (1-V)(1-W)\mathcal{E}}{1-D}$$

then gives $|H_p - 1| \leq 656Q^{-149/100} < 1024Q^{-149/100}$.

For $p \mid u$, $1 \leq j = v_p(u) \leq 5$, cancellation gives

$$H_p - 1 = \frac{R(1-Q^{-1}) - \eta(p)(Q-1)Q^{-x-w}V^{1_{j\leq 1}} + (1-V)J_j}{1-R}.$$

The middle term has exponent at most $-7/10$. The J_j exponents are

j	1	2	3	4	5
exponents	$-17/10$	$-3/4$	$-6/5, -1$	$-29/20$	$-3/2.$

There are at most two such terms. Thus $|H_p - 1| \leq 24Q^{-7/10}$, and the stated constant 32 is safe. These calculations include every row valuation; primes in the excluded set S have no factors here. The larger open region

$$\operatorname{Re} x > 74/100, \quad \operatorname{Re} w > 94/100, \quad \operatorname{Re} z > 31/200$$

retains respective exponents $-137/100$ and $-68/100$ and the same denominator control. This proves convergence on neighborhoods, including the boundary of the stated closed region. For any $\epsilon > 0$, the finite row product is $O_\epsilon(q_u^\epsilon)$ since, outside a fixed finite set, $1 + 32Q^{-68/100} \leq Q^\epsilon$ on that open neighborhood.

The source defines its full compensated correction as a finite sum over selected prime tuples. Each selected local replacement G_p has only $1-V, 1-R, 1-D$ denominators, and its unselected factors are the H_p just bounded. That formula therefore also extends holomorphically to this neighborhood for each fixed Z. No quantitative principal-error estimate for the compensated expression follows merely from holomorphy.

K.2 Principal factor, tails and nonvanishing

At $u = 1, w = 1, z = 1/6$, the source's P_p^* becomes $(1+v)(r-d)/(1-r)$ with the main-text definitions. This proves the displayed principal formula. Its numerator cannot vanish on $\operatorname{Re} s > 1/2$: $0 < v < 1$, $|d| < 1$, $|r| < 1$ and the strict convexity estimate apply. To check convergence, direct subtraction gives

$$H_p - 1 = \frac{-v^2 + r(1 - v + v^3) + d(v + v^2 - v^3) - rd}{(1-r)(1-d)}.$$

For $\sigma = \operatorname{Re} s > 2/3$, $|r| < 1/2$ and $|d| < 2/3$. The numerator is at most $v^2 + 2|r| + 2v|d|$, while the denominator reciprocal is less than 6. This gives

$$|H_p - 1| \le 12(Q^{-2} + Q^{-\sigma-1} + Q^{3-6\sigma}).$$

For completeness, the Eisenstein norm satisfies $a^2 - ab + b^2 \ge (a^2 + b^2)/2$. The norm-Euclidean ring is a PID; counting generators in $|a|, |b| \le 2\sqrt{X}$ bounds the number of ideals of norm at most $X \ge 1$ by $25X$. Dyadic decomposition consequently gives

$$\sum_{Np > P_0} (Np)^{-1-\delta} \le \frac{50}{1 - 2^{-\delta}} P_0^{-\delta} \qquad (\delta > 0).$$

For a fixed $\sigma_0 > 2/3$, use $\delta = \min(1, \sigma_0, 6\sigma_0 - 4) > 0$. Choosing P_0 so that the local defect sum is at most $1/4$ gives

$$|H_\eta(s) - 1| \le e^{1/4} - 1 < 1/3 \quad (\operatorname{Re} s \ge \sigma_0).$$

This choice is uniform in phases for that fixed half-plane. Its constants degenerate as $\sigma_0 \downarrow 2/3$; no uniform bound on the whole open half-plane is being inferred.

K.3 What the additional factor permits and prevents

Removing $1 - r$ leaves

$$A_p - 1 = -v^2 + \frac{(1-v^2)v(d-r)}{1-d}.$$

On every fixed $\operatorname{Re} s \ge \sigma_0 > 1/2$, this is $O_{\sigma_0}(Q^{-2} + Q^{-\sigma_0-1} + Q^{2-6\sigma_0})$. The product A is thus normally convergent and nonzero there, by local nonvanishing and summable defects. The remaining Euler factors are exactly those of $L_F^S(6s-3, \theta)$ with $\theta = \alpha^{-6}\eta^6$. Units have sixth power one, so the angular character is well-defined on ideals; a finite-order twist cannot cancel its nonzero angular type.

The stronger obstruction uses the following external classical inputs: the primitive completion Λ_θ is entire of order one, has gamma factor $\Gamma(s+3)$ times a nonzero exponential, obeys $\Lambda_\theta(s) = \epsilon_\theta \Lambda_{\bar\theta}(1-s)$, and $\Lambda_{\bar\theta}(\bar s) = \overline{\Lambda_\theta(s)}$. The nonvanishing Euler product holds on $\operatorname{Re} s > 1$. These are Hecke analytic theory, not GRH or Artin holomorphy assumptions; none is verified by the new Lean file. Tate's continuation/functional equation framework is [10, sections 4.4–4.5]. White's Theorem 2 [11] records the pure angular case, including the gamma shift; it is not a citation for arbitrary finite-conductor twists or a substitute for the stated order-one input.

Hadamard factorization with finitely many zeros would give $\Lambda_\theta(s) = e^{as+b}P(s)$. Along the positive real axis this has logarithmic modulus $\operatorname{Re}(a)\sigma + O(\log \sigma)$, whereas Stirling and $L(\sigma, \theta) \to 1$ give $\sigma \log \sigma + O(\sigma)$. Thus there are infinitely many zeros. Euler nonvanishing and the functional equation confine them to $0 \le \operatorname{Re} \rho \le 1$. Conjugation and the functional equation reflect a zero to $1 - \bar\rho$ for the same character, so some zero has $\operatorname{Re} \rho \ge 1/2$. Gamma is finite and nonzero there, as

are the deleted Euler factors; hence $L_F^S(\rho,\theta)=0$. The unique continuation $A(s)L_F^S(6s-3,\theta)$ then vanishes at $(\rho+3)/6$. Enlarging finite S cannot remove that zero. This proves the stated 7/12 obstruction under the explicit classical data; a replacement normalization/probe would need new analytic estimates.

The exact program `scripts/check_euler_correction.py` checks all exponents and evaluates the identities at primary prime representatives $-2-3\omega$ and $1-3\omega$, of norms 7 and 13, using only $\mathbb{Q}(\sqrt{-3})$ arithmetic. Lean checks scalar cancellation, convex local nonvanishing, genuine complex infinite-product components under explicit summability hypotheses, and shifted-zero transport. The arithmetic convergence estimates, source Gauss identification, multivariable holomorphy and classical Hecke completion theory remain prose/external dependencies. This does not establish the missing compensated contour bounds, mean-square estimates or GRH.

L Principal multiplier, weighted slots and the extended residue comparison

The source passages are lines 6173–6328, 7004–7044, 9050–9097 and 15548–15669 of the pinned September paper. We retain its displayed scalar/high identities and Mellin/reciprocal contracts as stated inputs.

L.1 Local errors and the scalar normalizer

Its exact cancellation for G_p+H_p consists of a bounded prefactor times

$$(1-Q^{-w})\{V/(1-V)-D\}+(\overline{\eta(p)}Q^{x}+1-Q^{-w})\mathcal{E}_p.$$

The first term costs $Q^{-6z_r}+Q^{-x_r}$. Multiplying the previous residual bound by Q^{x_r} costs $Q^{4-5x_r-6z_r}+Q^{1-w_r-6z_r}$; unamplified residual terms are smaller. Division by the uniformly nonzero H_p therefore gives

$$|\mathcal{B}_p+1|\ll Q^{-\kappa},\qquad \kappa=\min(x_r,6z_r,5x_r+6z_r-4,w_r+6z_r-1)\geq 37/50.$$

It follows that $G_p=O(1)$ on the enlarged principal region. On a fixed real box, an annular slot contributes at most $O(P_i^{z_r})$ by ideal counting, since its weight is Q^{z-1}. The unselected product has a bounded positive majorant. Thus the full tuple is $O(Z^{\ell z_r})$ uniformly in heights, as required. This is not a constant uniform over unbounded z_r.

For the prime normalizer, the fixed-extension consequence of [12, Theorem 1.1] gives $\pi_{1_T}(x)\sim \mathrm{Li}(x)/|T|$. Any exceptional zero is fixed below 1, so its secondary term is asymptotically smaller; it need not be ruled out. Stieltjes integration against the fixed annular weight gives

$$S_i(Z)\sim \frac{P_i^{1/6}}{|T|\log P_i}\int_0^\infty W_i(y)y^{-5/6}\,dy>0.$$

A further fixed finite deletion eventually leaves the window entirely. The normalizer consequently has the signed asymptotic

$$A_T(Z)\sim \frac{c_T}{(\log Z)^K},\qquad c_T=\frac{(-1)^K}{|T|^K}\prod_i\left(\ell_i^{-1}\int_0^\infty W_i(y)y^{-5/6}\,dy\right)\neq 0.$$

This proves the inverse subpower bound. For example, after setting $u=\log Z$, the exponential series shows that $e^{\epsilon u}$ eventually dominates the fixed power u^K. Thresholds may depend on the fixed ray and excluded set; the power-saving exponents do not.

At the residue write $a_p = W_i(Q/P_i)Q^{-5/6} \geq 0$ and $\varepsilon_p = \mathcal{B}_p + 1$. Then

$$\sum_p a_p \mathcal{B}_p = -S_i(1+\rho_i), \qquad \rho_i = -\frac{\sum_p a_p \varepsilon_p}{S_i}, \qquad |\rho_i| \ll P_i^{-37/50}.$$

The sign of ρ_i matters. Nonnegative individual weights permit division by their positive total without cancellation. A merely positive sum of signed weights would not give this inequality. For a fixed finite set of K errors bounded by $\delta \leq 1$,

$$\left| \prod_i (1+\rho_i) - 1 \right| \leq (2^K - 1)\delta.$$

Induct by writing $(1+\rho_i)P - 1 = (1+\rho_i)(P-1) + \rho_i$ and using $|1+\rho_i| \leq 2$. Thus the displayed residue identity holds with $\mu = (7/10)\ell_{\min}$, where $\ell_{\min} > 0$ is fixed before the target. The same signed A_T is used in every normalization of this probe.

L.2 Contour powers and the perturbed geometry

Use the inherited exact high representation for the same physical probe, the rapid integrated and trace tail contracts for the Mellin tests, and polynomial reciprocal control to the right of β_*. For $3/4 \leq \sigma_0 < \beta_*$, the principal paths can be moved to

$$\operatorname{Re} s = \beta_* + e, \quad \operatorname{Re} w = 1 + e, \quad \operatorname{Re} z = 1/6 + e$$

inside the enlarged domain. Move w left to $19/20$; in its residue move z left to $33/200$. Only the two scalar zeta poles are crossed. The reciprocal remains on the global line; at a principal pole its zero causes no additional residue. Horizontal joins use the trace estimates in the independent height coordinates $(\Im(s+z), \Im z, \Im w)$. After the w residue, use the corresponding two-dimensional kernel. Arithmetic height orders are fixed first; later external tail orders increase only the decay requested from the tests.

The raw exponent is

$$l_x/2 + \operatorname{Re} s - 1 + h \operatorname{Re} z + l_y(\operatorname{Re} w - 1), \qquad h = 1 - l_x + \ell.$$

On the unresidued w integral it is $C(\beta_*) + (1+h)e - l_y/20$; on the leftover z integral it is $C(\beta_*) + e - h/600$. At the double residue the normalization gives the main signal times $1 + \mathcal{R}_{\eta,Z}(s)$. Its error costs $Z^{C(\beta_*)+e-\mu+\epsilon}$ on the entire global line, using the Gaussian and reciprocal bound. Only the main signal moves right to $\operatorname{Re} s = 2$; no continuation of the relative error is used. The factor $1/6$ is already in c_S from the residue of $\zeta_F^S(6z)$. The normalizer's inverse is absorbed in the allocated ϵ. This proves the three remainder powers stated in the main text, conditional on the explicit inherited analytic contracts.

For $t = 10^{-5}$, the perturbed parameters are

$$\ell = 1/6 + t, \quad l_x = 17/48 - t/2, \quad l_y = 23/48 - t/2, \quad h = 13/16 + 3t/2.$$

They have $C(s) = s - 11/16$ and $\sigma_t = 349999/400000 > 3/4$. Put $m_w = l_y/20$, $m_z = h/600$, and choose

$$0 < e = \epsilon \leq \min\left(\frac{1}{10000}, \frac{m_w}{8(1+h)}, \frac{m_z}{8}, \frac{\mu}{8}\right).$$

Each total error cost is at most one quarter of its corresponding margin, so all three remainders have a common positive saving fixed before the target. Nonprincipal contour, energy and frozen-capacity row-count extensions remain separate obligations.

L.3 A pointwise obstruction at $3/5$

At the principal residue put $N = 1 - vr - (1-v)d$. In addition to the main-text multiplier identity, exact algebra gives

$$\frac{\mathcal{B}_p + 1}{r/d} - 1 = \frac{-v(1-r) + (1-v)^2 d(1-d/r)}{N}.$$

For fixed $1/2 < \sigma \le 3/5$ and $Q \ge 4$, $|N| \ge 1/4$, $|r| < 1$ and $|d| < 1/2$. Hence the relative error is at most

$$8Q^{-1} + 4Q^{-\sigma} + 4Q^{4\sigma-3} \longrightarrow 0.$$

Since $|r/d| = Q^{3-5\sigma}$, a pointwise negative-power bound around -1 fails in this range, uniformly in phases and heights.

For an exact arithmetic endpoint example, let $\eta = 1$ and let the rational prime $p \equiv 2 \pmod 3$ be inert. Its prime ideal has norm $Q = p^2$ and primary generator $-p$, so $\alpha^{-6} = 1$. At real $s = 3/5$, $r = Q^{3-18/5} = Q^{-3/5} = d$. Thus $P_p^* = 0$ and $\mathcal{B}_p = -Q^{-1}$. Given a fixed ray modulus $\mathfrak{m}$, choose a positive integer M with $M\mathcal{O} \subset \mathfrak{m}$. Primes $p \equiv -1 \pmod{3M}$ have primary generator $-p \equiv 1 \pmod{\mathfrak{m}}$ and occur infinitely often by the classical fixed-progression prime theorem. They obey the identity-ray restriction. Nevertheless their ideals have count $O(\sqrt{P})$ up to norm P, a zero-density subset of the prime ideals. This supplies no exclusion of cancellation in the full slot average.

The exact checker `scripts/check_principal_residue.py` verifies the multiplier and slot identities, rational gaps and illustrative principal loss allocations. Its slot counts do not certify row-count admissibility. The six Lean statements verify scalar identities, actual complex weighted normalization and finite-product error transfer. Prime counting, the arithmetic endpoint identification and all contour integrations remain prose/external inputs. No new zero-free half-plane follows from this principal interface alone.

M Research tasks with mathematical acceptance criteria

Track A: repair the determinant interface. Start with the exact diagonal Euler transfer on $\operatorname{Re} s > 1$. Determine which regularized operator determinant could continue the reciprocal Euler product without poles or zeros being silently exchanged. Prove a relation to a self-adjoint operator before replacing the transfer by its Hermitian part. Acceptance requires an identity of functions, not similarity of zero plots. The displayed 2×2 example rejects unrestricted determinant preservation immediately.

Track B: trace norms and normal families. Choose a connected domain containing an Euler-product overlap and the critical strip, after removing the known poles in the target. Supply estimates uniform in N on every compact set. Either construct a trace-class holomorphic family with uniform trace-norm convergence or prove the hypotheses of Proposition 2.3. The divergence in Theorem 3.3 prevents an absolute prime-sum shortcut. Any cancellation estimate must be independently proved without a zero-free region equivalent to the desired conclusion.

Track C: positivity and canonical products. Compare the exact spectral quadratic form with Weil positivity and the Li coefficients for the chosen L-function family. First state the test-function space and regularization, then prove equality of the forms. Positivity of a finite matrix or of finitely many Li coefficients is insufficient; a uniform infinite-family statement is required. Canonical products provide a convergence benchmark, but the preceding converse explains the circularity risk of using zeros as input.

Track D: make formalization match mathematics. The standalone Lean companion checks exact integer and finite-matrix obstructions. A separate mathlib layer now verifies complex Euler-factor nonvanishing, a local boundary-convergence theorem, and holomorphic Hermitian rigidity. It does not yet formalize the Dirichlet partial-sum convergence or an arithmetic self-adjoint operator. Further formalization must use the actual standard L-functions and explicit analytic domains. An abstract theorem of the form "if an assumed bridge implies GRH, then GRH" is logically valid but would not verify the bridge. Print every theorem's axioms and review its full type and dependencies, not merely its name.

N Computational examples and reproducibility

The Python checks use integers and `fractions.Fraction`, with no random choices, model weights, scientific datasets or floating-point zero finding. They reproduce the Iwahori witness at $\pi = 2$, the determinant change $2 \to 1$, the parity obstruction, and the vanishing diagonal model. The Lean statements verify selected finite calculations and the parity argument in the kernel of a pinned Lean 4 toolchain. Neither executable claims to verify the full complex-analytic arguments in this article.

Both editions share the same core source. The longer edition adds these calculations and research interfaces; extra exposition does not strengthen an unproved hypothesis into a theorem. The source manifest and audit coverage record distinguish full text review, focused review, and unverified dependencies. Publication readiness also requires independent mathematical review, confirmed authorship, and journal-specific preparation.

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